

Normalization Process

Hotel Management System

Group 7

ADDIS GRAND HOTEL

1. Unnormalized Form (UNF)

Initially, all hotel data was stored in a single large table with many problems:

Unnormalized Table:

Bookin gID	Guest Name	Phone	Email	Room Numbers	Check- in	Check- out	Total Amount	Room Type	Price	Ordered Items
B001	Biruk Molla	091213...	biruk@gmail.com	A-101, B- 205	2026- 04-20	2026- 04-25	8500	Double	1200	Chicken, Fries, Soft Drink

Major Problems:

- Repeating groups (multiple rooms and items)
- Multi-valued attributes
- High data redundancy
- Update, insert, and delete anomalies

2. First Normal Form (1NF) → Before → After

Improvement: Removed repeating groups and made every attribute atomic (single value).

1NF Table:

BookingID	Guest Name	Phone	Email	Room Number	Check-in	Check-out	Total Amount	Room Type	Price Per Night
B001	Biruk Molla	091234..	biruk@gmail.com	A-101	2026-04-20	2026-04-25	4250	Double	1200
B001	Biruk Molla	091234..	biruk@gmail.com	B-205	2026-04-20	2026-04-25	4250	Deluxe	1200

3. Second Normal Form (2NF) → Before → After

Problem: Partial Dependency (Room Type and Price depend only on Room, not on Booking).

Solution: Split into two tables.

- **Booking Table:** BookingID, Guest Name, Phone, Email, Check-in, Check-out, Total Amount
- **Room Table:** Room No, Room Type, Price Per Night

4. Third Normal Form (3NF) → Before → After

Problem in 2NF: Transitive dependency (Guest Name, Phone, Email depend on Guest, not directly on Booking).

Solution: Further splitting, Separated Guest information.

Final 3NF Tables:

- **Guest** (guest_id PK, full_name, phone, email, address, nationality)
- **Booking** (booking_id PK, guest_id FK, check_in_date, check_out_date, total_amount, status)
- **RoomType** (type_id PK, type_name, price_per_night, max_occupancy)
- **Room** (room_id PK, room_number, type_id FK, status)

5. Boyce-Codd Normal Form (BCNF)

All determinants are now candidate keys. No anomalies remain.

Final Normalized Schema:

1. **RoomType** (type_id, type_name, price_per_night, max_occupancy, description)
2. **Room** (room_id, room_number, type_id, status, floor_number)

3. **Guest** (guest_id, full_name, phone, email, address, nationality, id_number)
 4. **Booking** (booking_id, guest_id, check_in_date, check_out_date, total_amount, status)
 5. **BookingRoom** (booking_id, room_id) – Junction table
 6. **Payment** (payment_id, booking_id, amount, payment_date, payment_method)
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Summary Box

Benefits of Normalization up to BCNF

- Eliminated data redundancy
- Removed update, insert, and delete anomalies
- Improved data integrity and consistency
- Easier maintenance and future expansion
- Better performance for queries and reporting

Conclusion: The database is now in **Boyce-Codd Normal Form (BCNF)**, which ensures high data quality and efficiency for the Addis Grand Hotel Management System.